

Configuring a Virtual Network in Hyper-V

Student Name: Mubeen Khan

Lab Environment: CloudLabs VM (Browser-based)

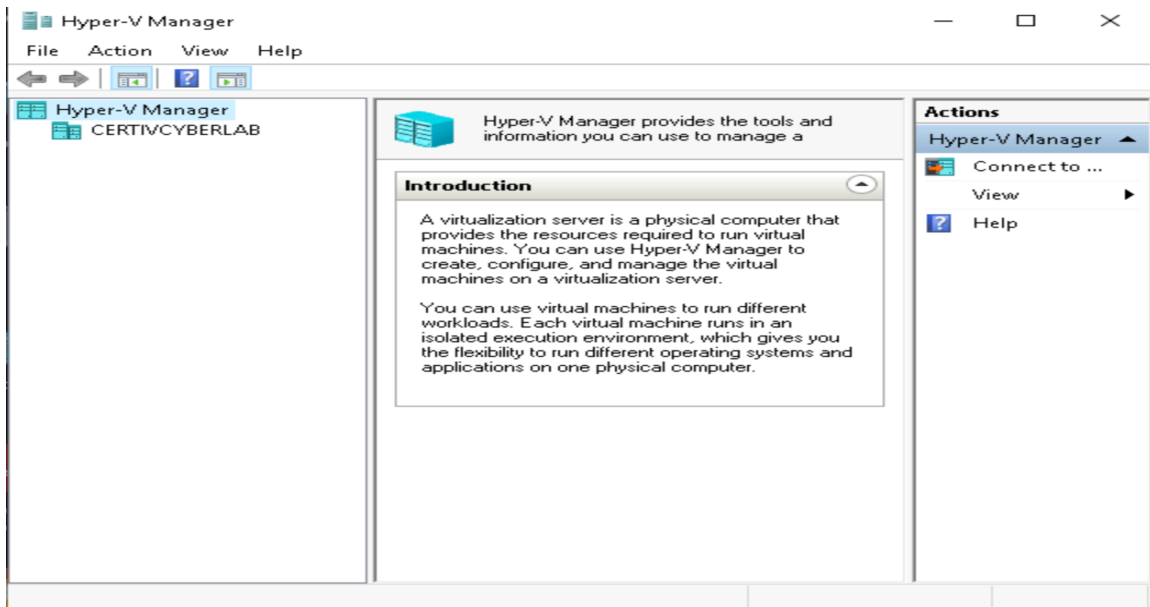
Lab Objective

This lab is part of my Certificate IV in Cyber Security from TAFE NSW. The objective was to configure a virtual network using Hyper-V, including NAT, DHCP, and Routing and Remote Access, and to validate connectivity using a nested virtual machine.

Task 1: Setting up Virtual NAT Network

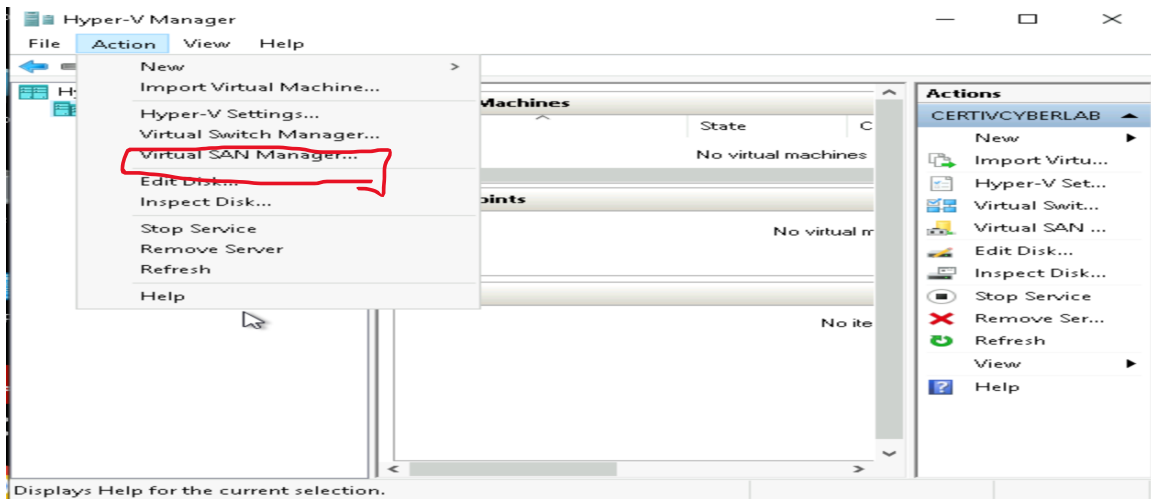
Step 1: Creating a Virtual Switch

Open Hyper - V manager

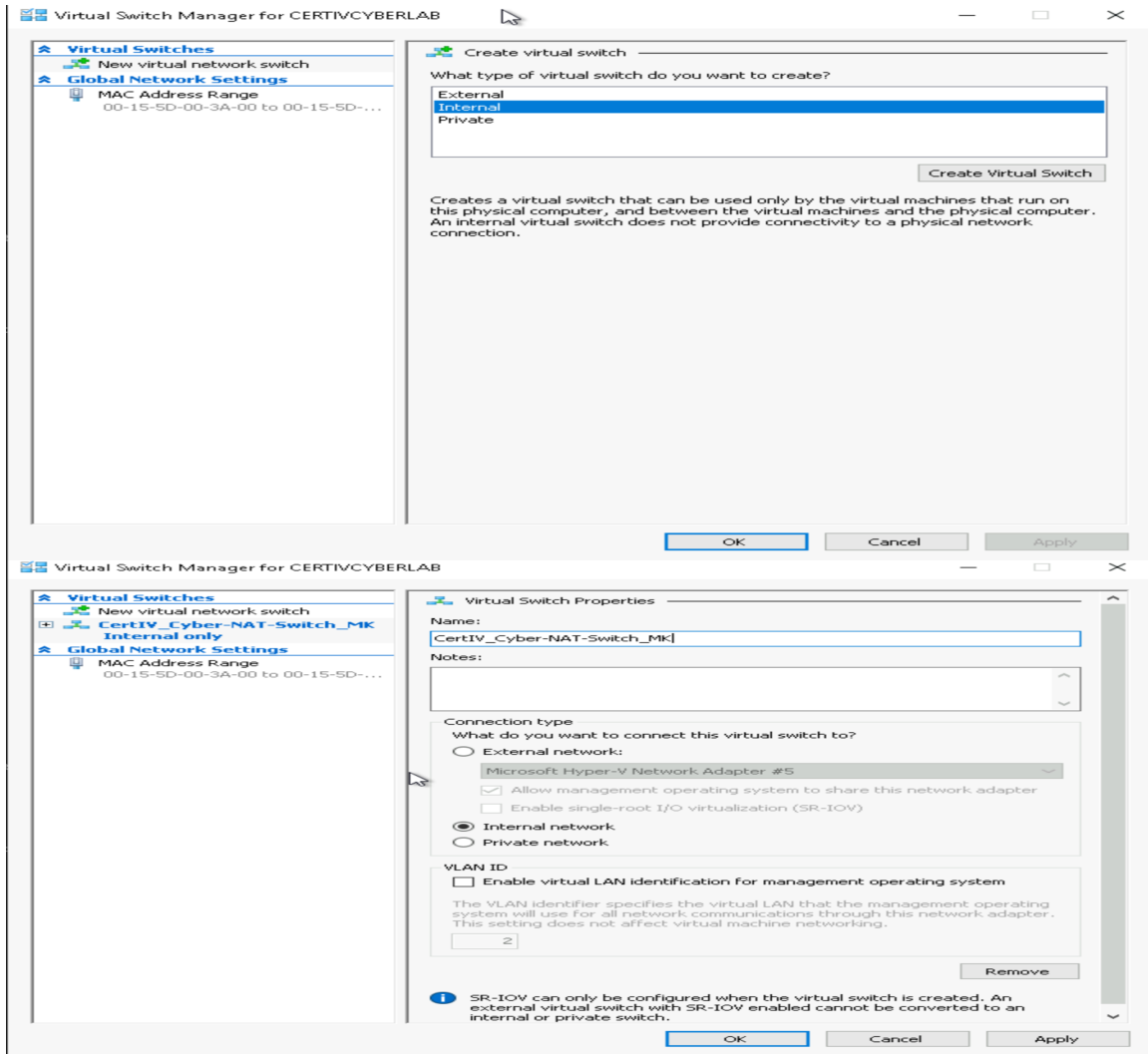


Select your server (in my case it is CERTIVCYBERLAB)

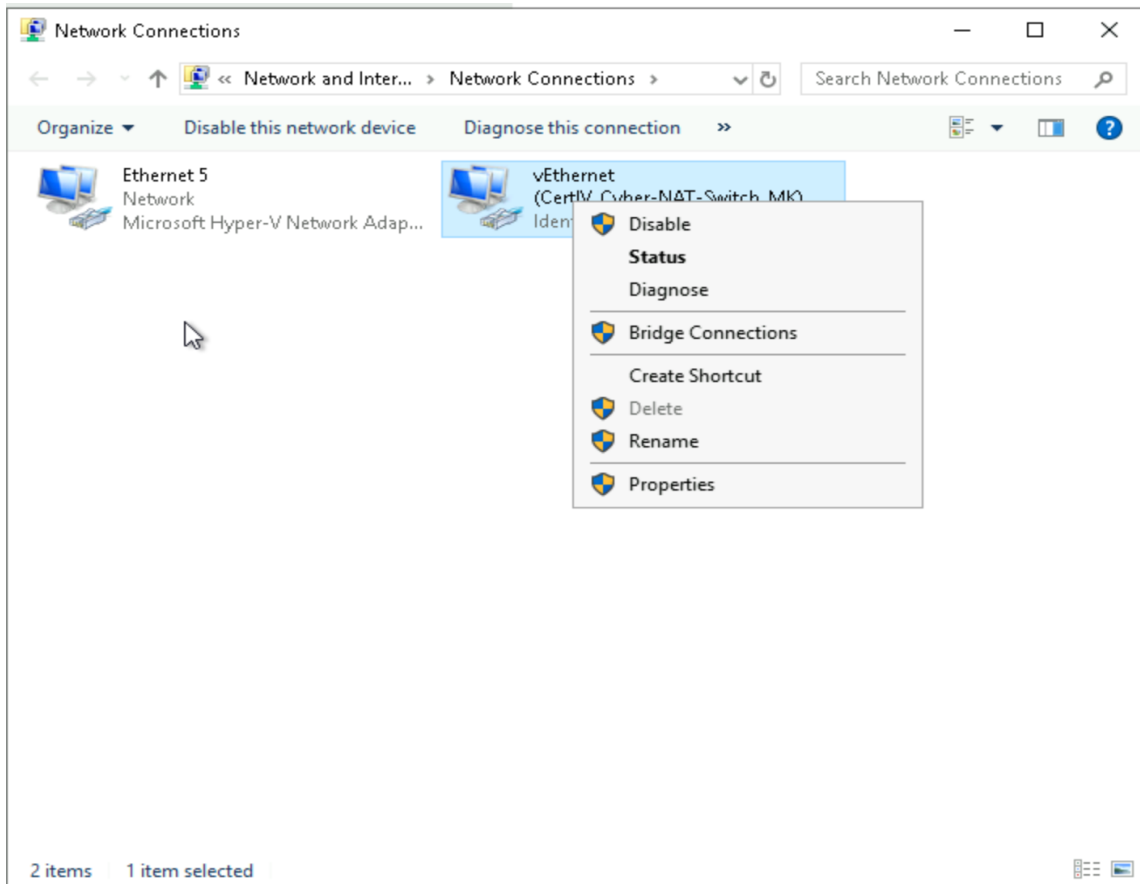
Then open the Virtual Switch Manager which is under the "Actions"



Create Internal Switch → Click on internal to select it and then click create virtual switch

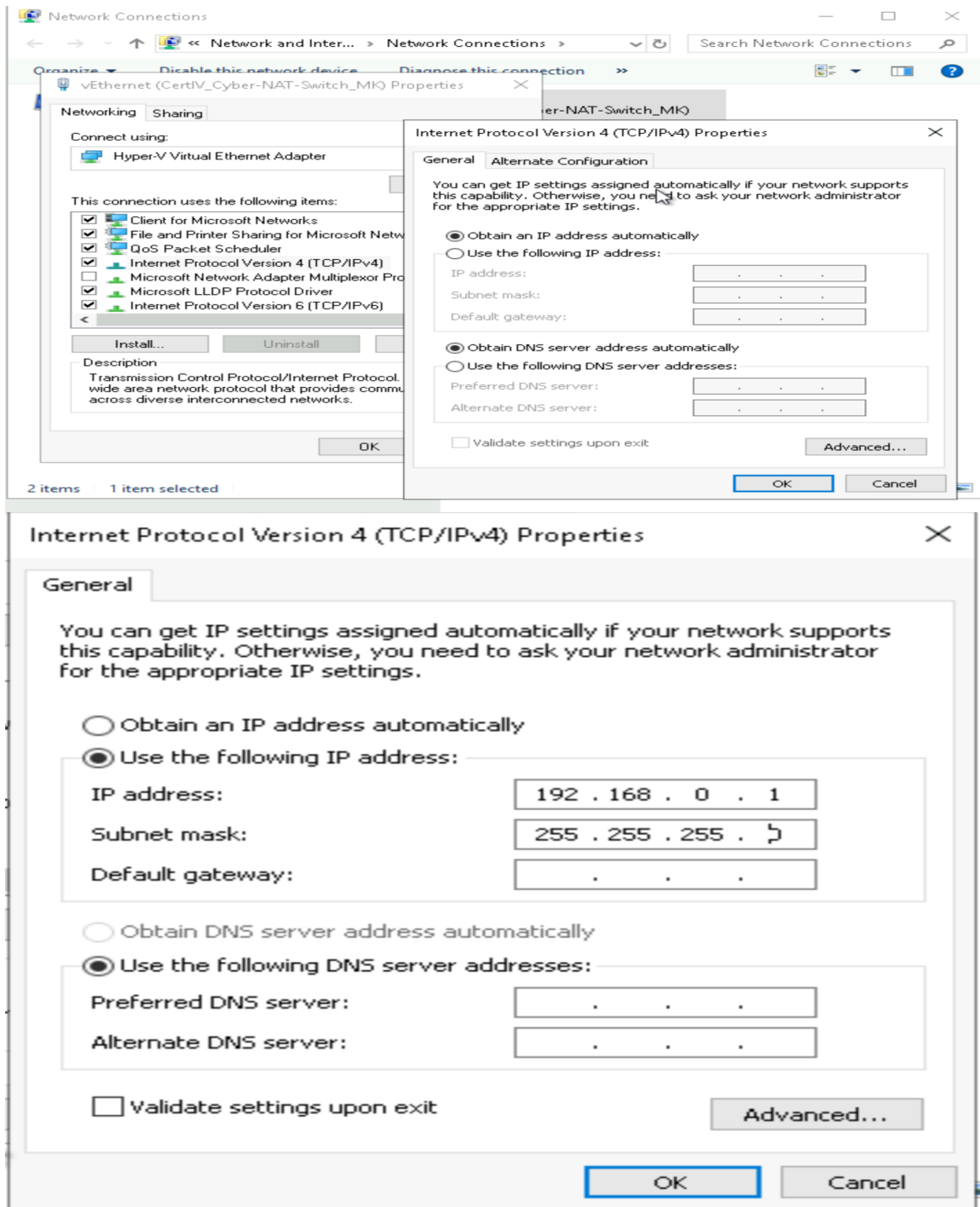


Verify the Virtual Switch was created by right-clicking on the network icon → open network & internet settings → Ethernet → Change adapter options, you should be able to see vEthernet (in my case CertIV-Cyber-NAT-Switch_Mk)



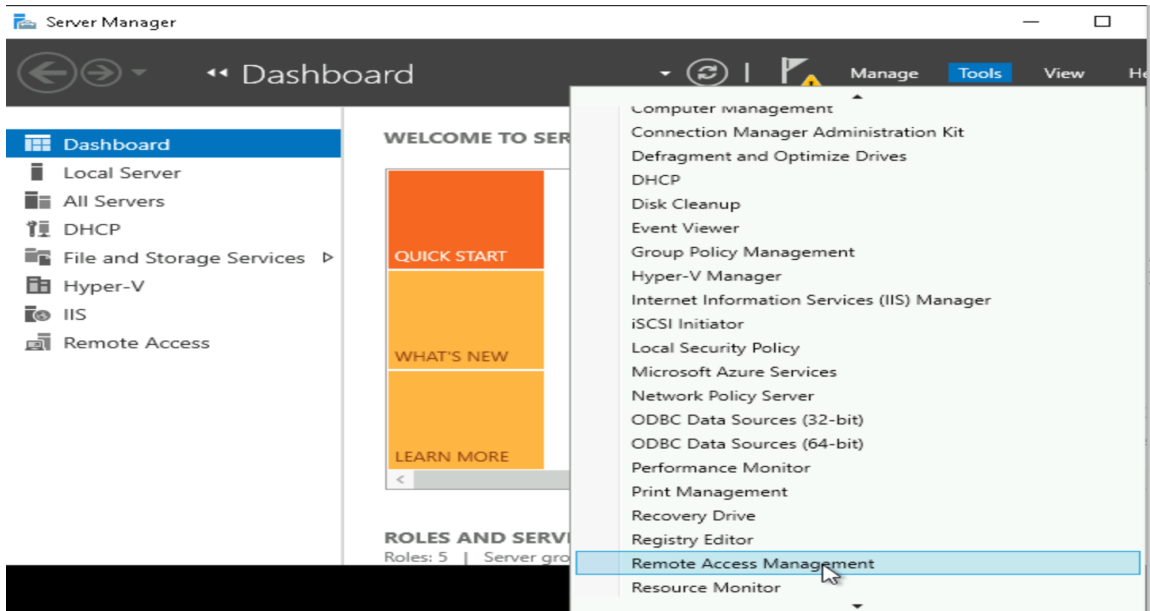
Setting up the IP Address on the Virtual Switch

Right-click vEthernet → Properties → Double click Internet Protocol Version 4 → Set up the IP address → OK → Close



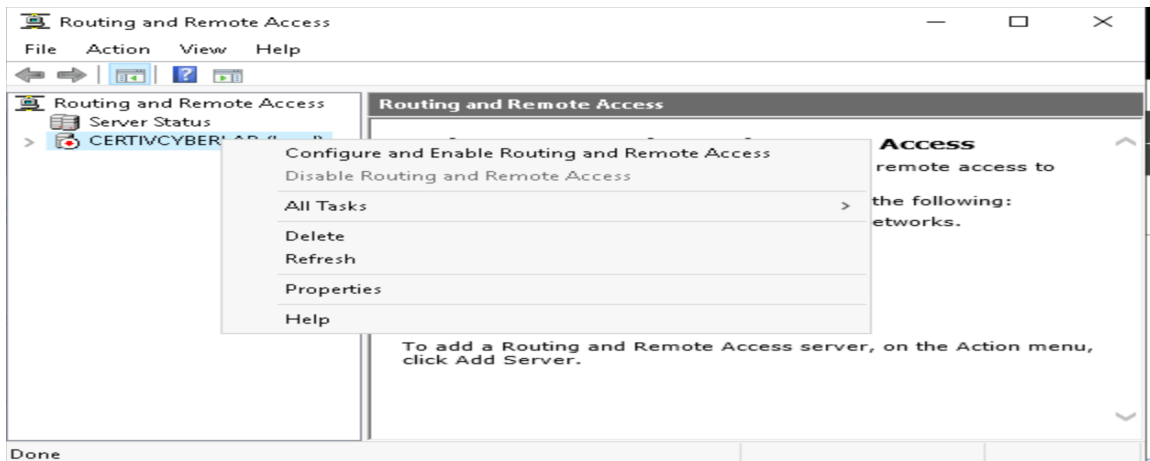
Step 2: Configuring Routing and Remote Access

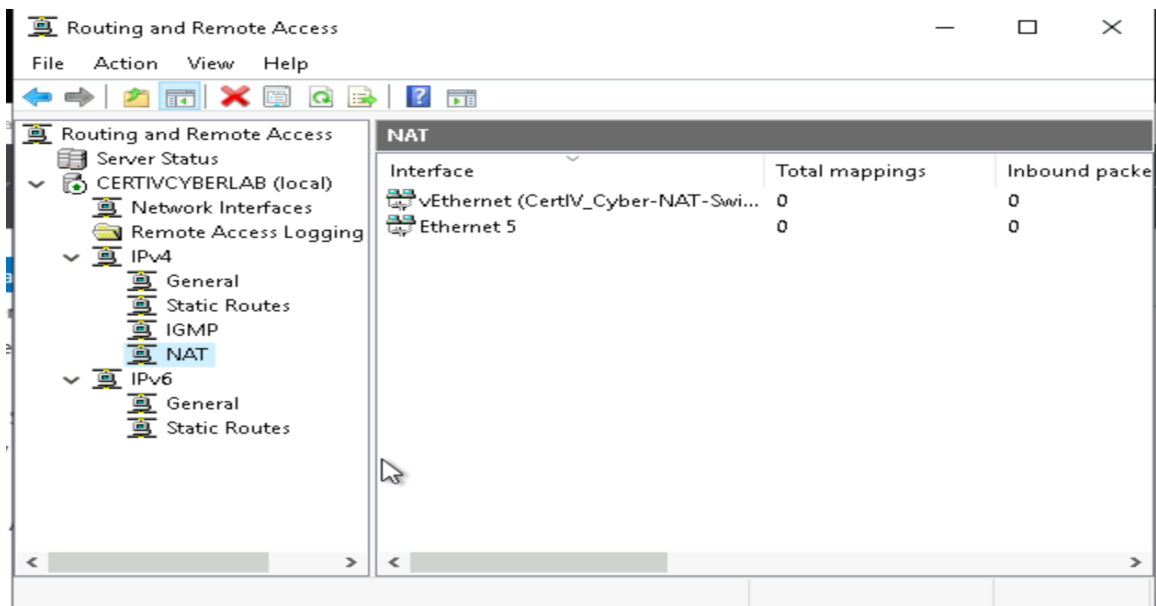
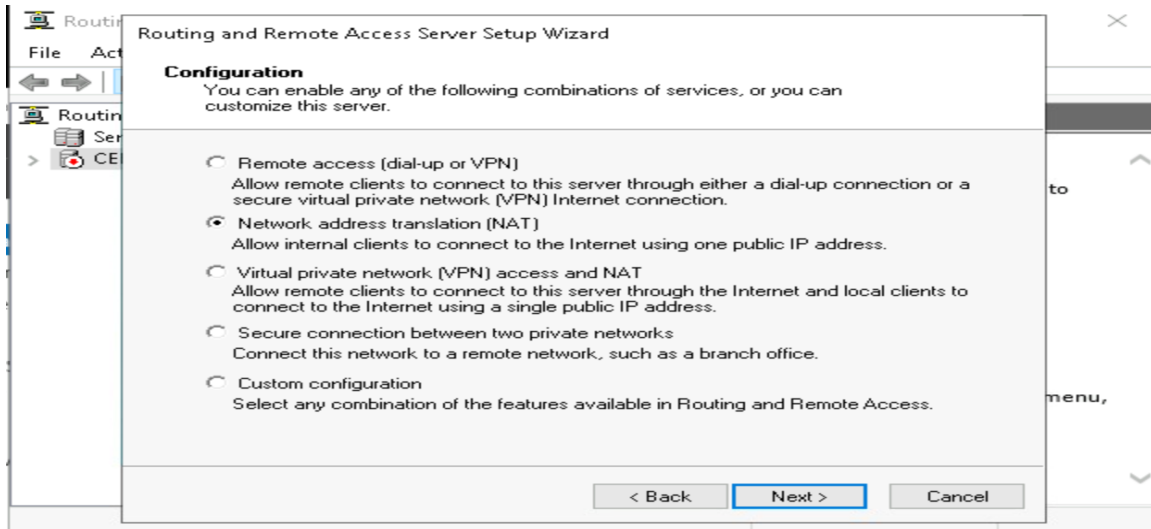
Open Server Manager, click Tools → Routing and Remote Access



Configure and enable Routing and Remote Access, right-click on your server name →
 Configure and Enable Routing and Remote Access → Next → Choose NAT configuration →
 Select public internet interface → choose Ethernet 2 → Next → Finish → OK

Set Ethernet 2 or 5 as public and set up Virtual Switch (vEthernet) as Private

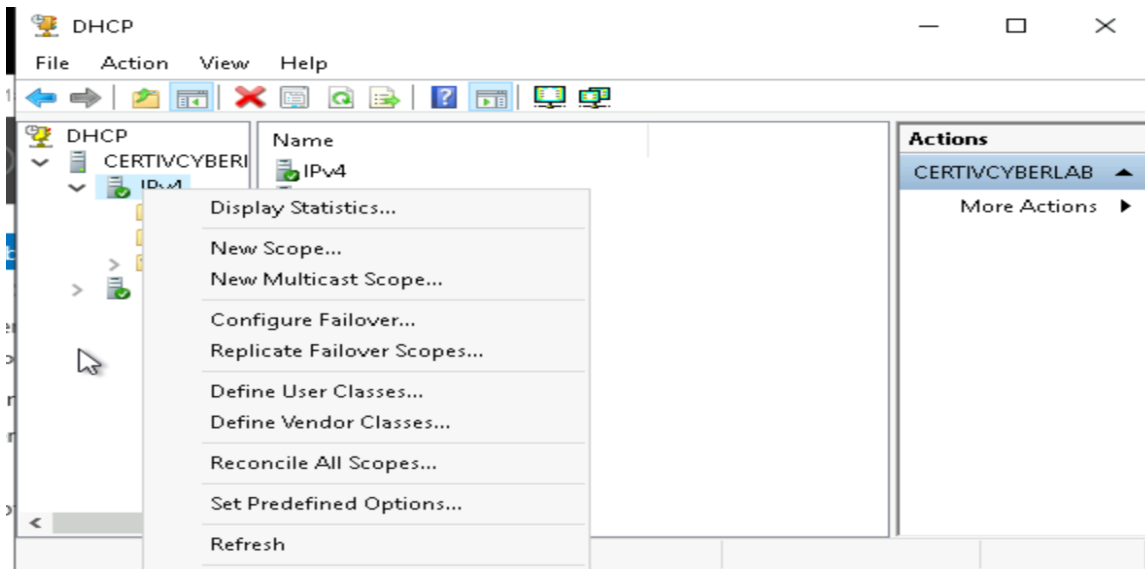




Step 3: Creating the DHCP Scope

Open DHCP (Server Manager → Tools → DHCP)

Start New Scope, expand your server → Right-click IPv4 → New Scope → Next → Name the Scope → Next



New Scope Wizard

Scope Name
You have to provide an identifying scope name. You also have the option of providing a description.

Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.

Name:

Description:

< Back **Next >** Cancel

Set the IP Address range → Skip exclusions → Next → Skip Lease duration → Next → Configure DHCP options (select yes) → Next

New Scope Wizard

IP Address Range

You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address: 192 . 168 . 0 . 100

End IP address: 192 . 168 . 0 . 200

Configuration settings that propagate to DHCP Client

Length: 24

Subnet mask: 255 . 255 . 255 . 0

< Back Next > Cancel

Set default gateway → Click Add → next → Set DNS server → Click Add → Next → No changes to WINS Server → Next → Activate scope → Next → Finish

New Scope Wizard

Router (Default Gateway)

You can specify the routers, or default gateways, to be distributed by this scope.



To add an IP address for a router used by clients, enter the address below.

IP address:

Add

192.168.0.1

Remove

Up

Down

< Back

Next >

Cancel

New Scope Wizard

Domain Name and DNS Servers

The Domain Name System (DNS) maps and translates domain names used by clients on your network.



You can specify the parent domain you want the client computers on your network to use for DNS name resolution.

Parent domain:

To configure scope clients to use DNS servers on your network, enter the IP addresses for those servers.

Server name:

Resolve

IP address:

Add

168.63.129.16

Remove

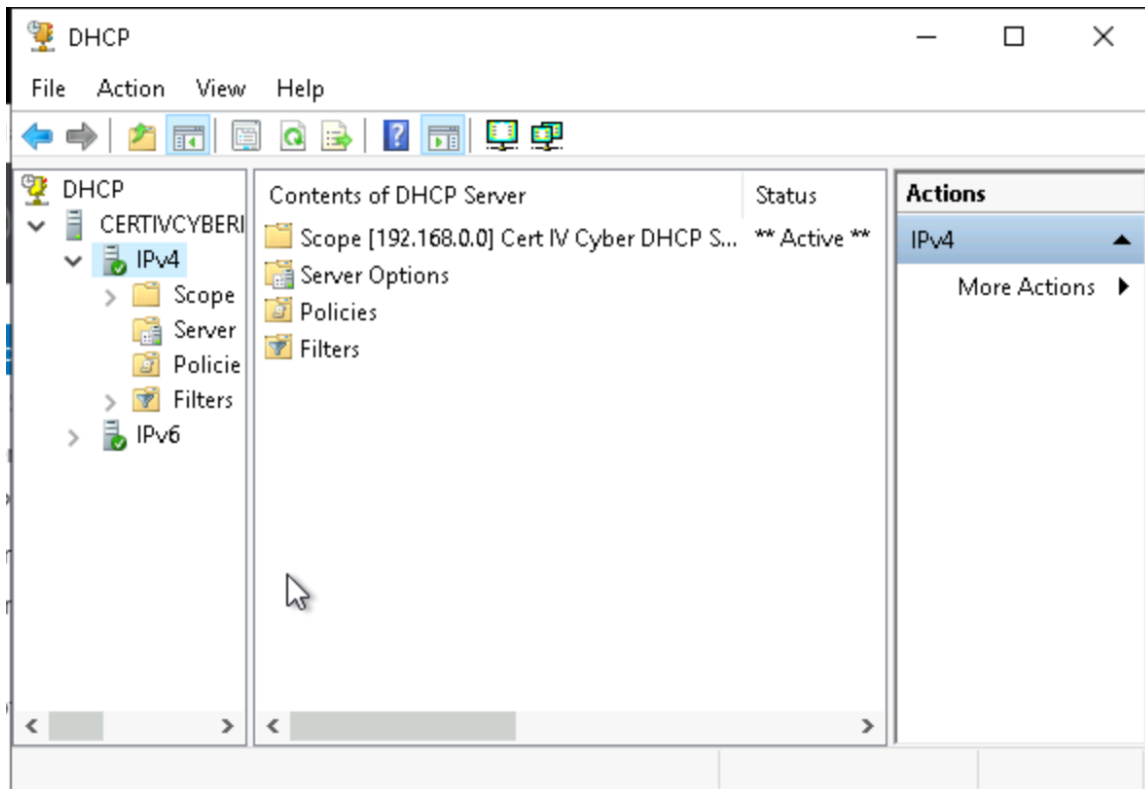
Up

Down

< Back

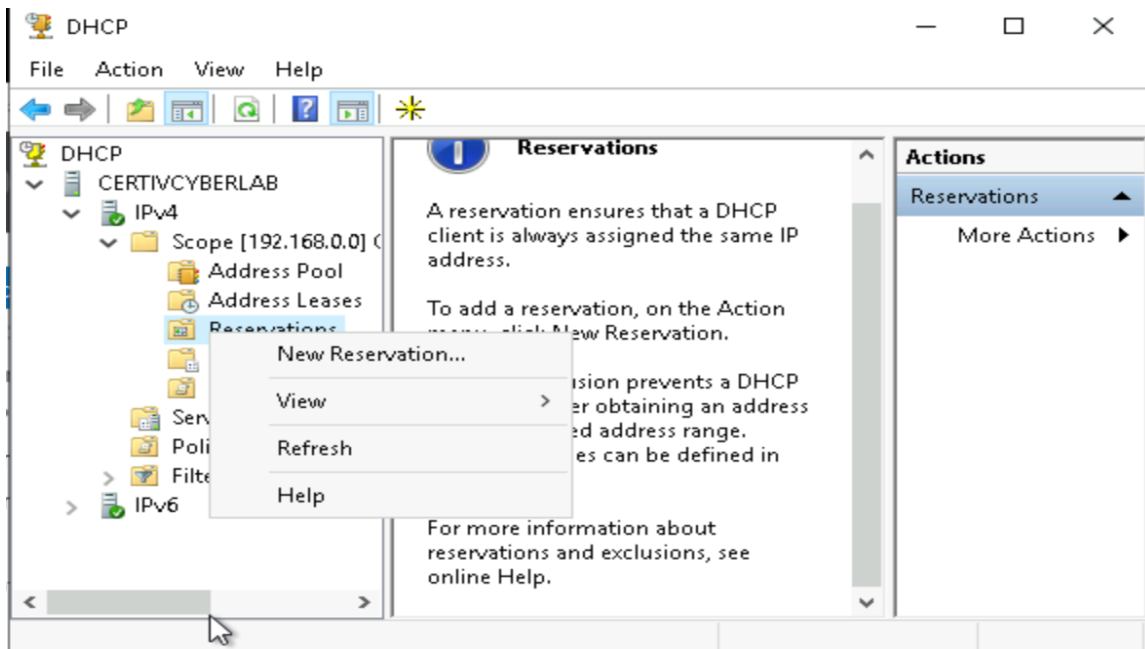
Next >

Cancel



Step 4: Creating IP Reservations

Navigate to Reservations, In DHCP console, expand your server → IPv4 → Your scope → Right-click reservations → New Reservations.



Fill in the Reservation Details → Add → Close

New Reservation

Provide information for a reserved client.

Reservation name: Reservation for Metasploitable VM

IP address: 192 . 168 . 0 . 50

MAC address: 00155d000703

Description: Reservation for Metasploitable VM

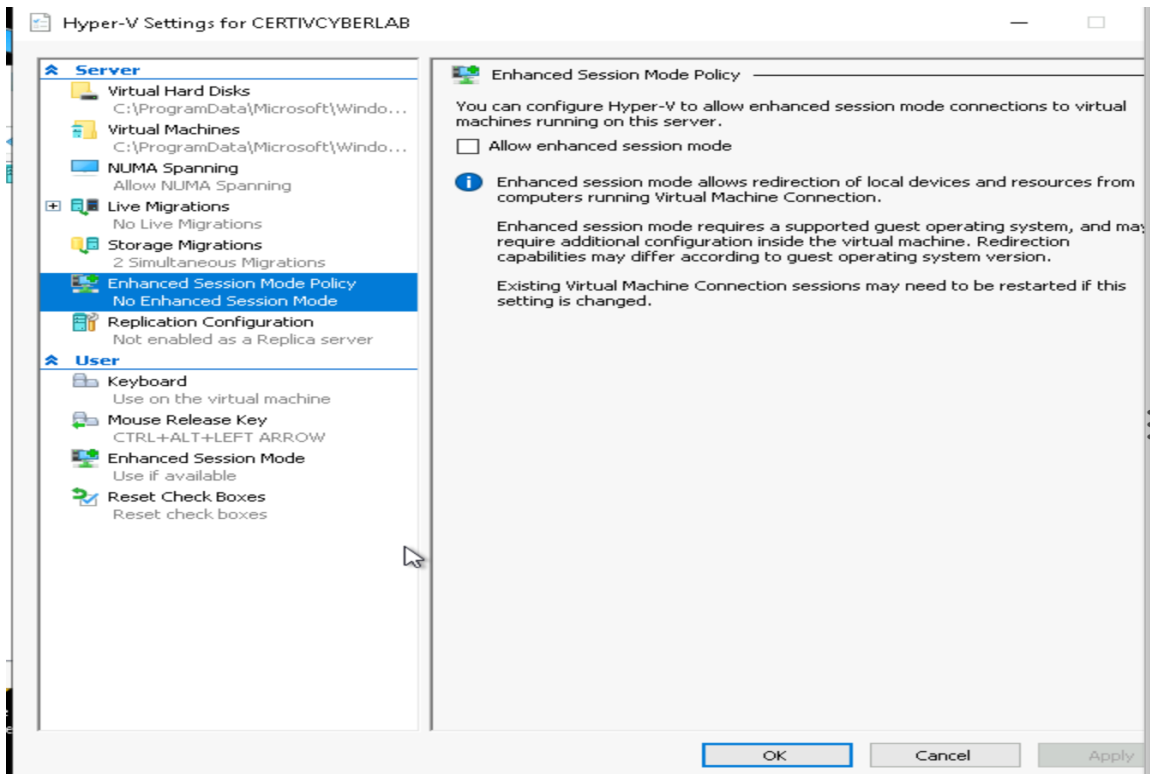
Supported types

- ☒ Both
- ☐ DHCP
- ☐ BOOTP

Add Close

Step 5: Enabling Enhanced Session Mode

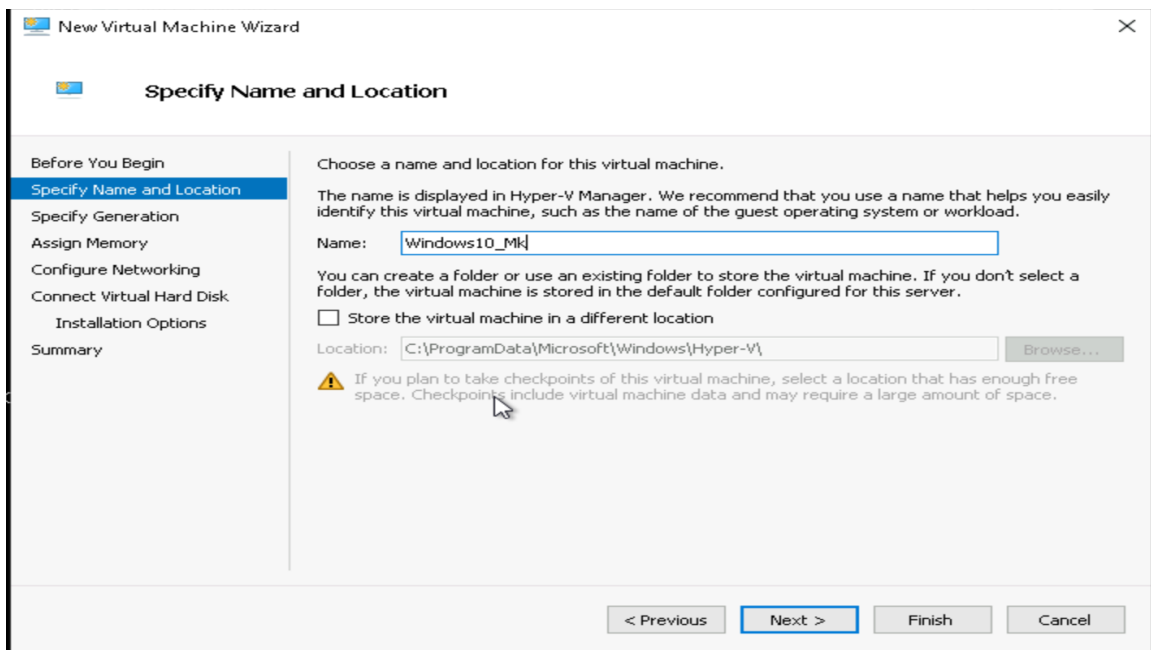
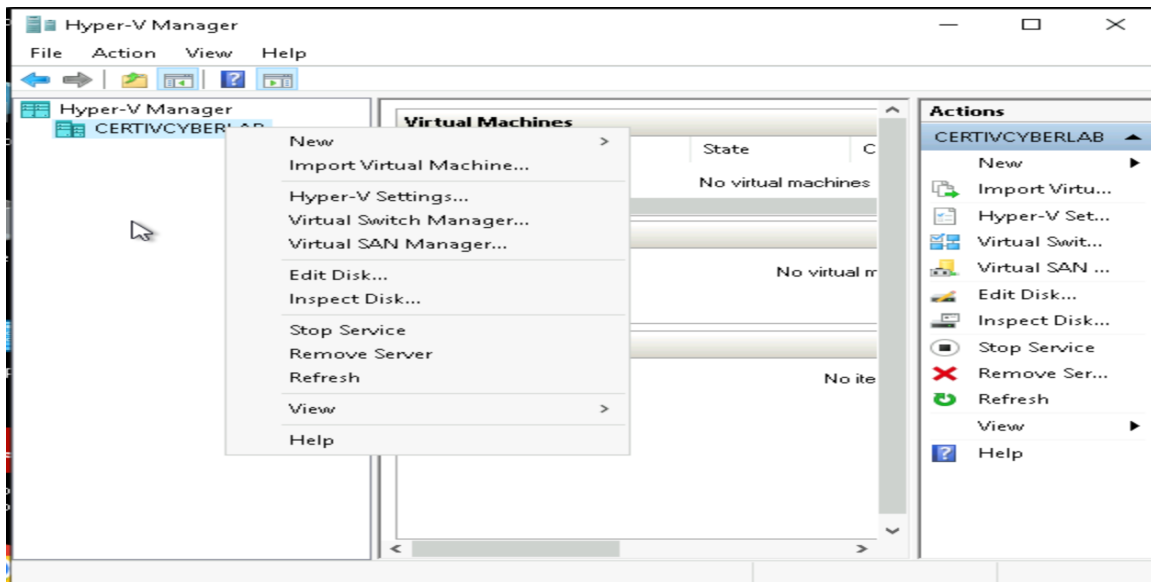
Open Hyper-V manager → Hyper-V settings → Click Enhanced Session Mode Policy → Apply → Ok



Task 2: Testing Network Configuration

Step 1: Creating a Nested Virtual Machine

Open New Virtual Machine wizard, Actions → New → Virtual Machine → Next → Name your VM → Next → Generation 2 → Next → Assign startup memory (4096) → Uncheck dynamic memory → next → Select NAT switch → Next → Select use and existing virtual hard disk → Go to your virtual disk windows 10 → Open → Next → Finish



New Virtual Machine Wizard

Specify Generation

Before You Begin

Specify Name and Location

Specify Generation

Assign Memory

Configure Networking

Connect Virtual Hard Disk

Installation Options

Summary

Choose the generation of this virtual machine.

☐ Generation 1

This virtual machine generation supports 32-bit and 64-bit guest operating systems and provides virtual hardware which has been available in all previous versions of Hyper-V.

☒ Generation 2

This virtual machine generation provides support for newer virtualization features, has UEFI-based firmware, and requires a supported 64-bit guest operating system.



Once a virtual machine has been created, you cannot change its generation.

[More about virtual machine generation support](#)

< Previous

Next >

Finish

Cancel

New Virtual Machine Wizard

Assign Memory

Before You Begin

Specify Name and Location

Specify Generation

Assign Memory

Configure Networking

Connect Virtual Hard Disk

Installation Options

Summary

Specify the amount of memory to allocate to this virtual machine. You can specify an amount from 32 MB through 12582912 MB. To improve performance, specify more than the minimum amount recommended for the operating system.

Startup memory: MB

☐ Use Dynamic Memory for this virtual machine.



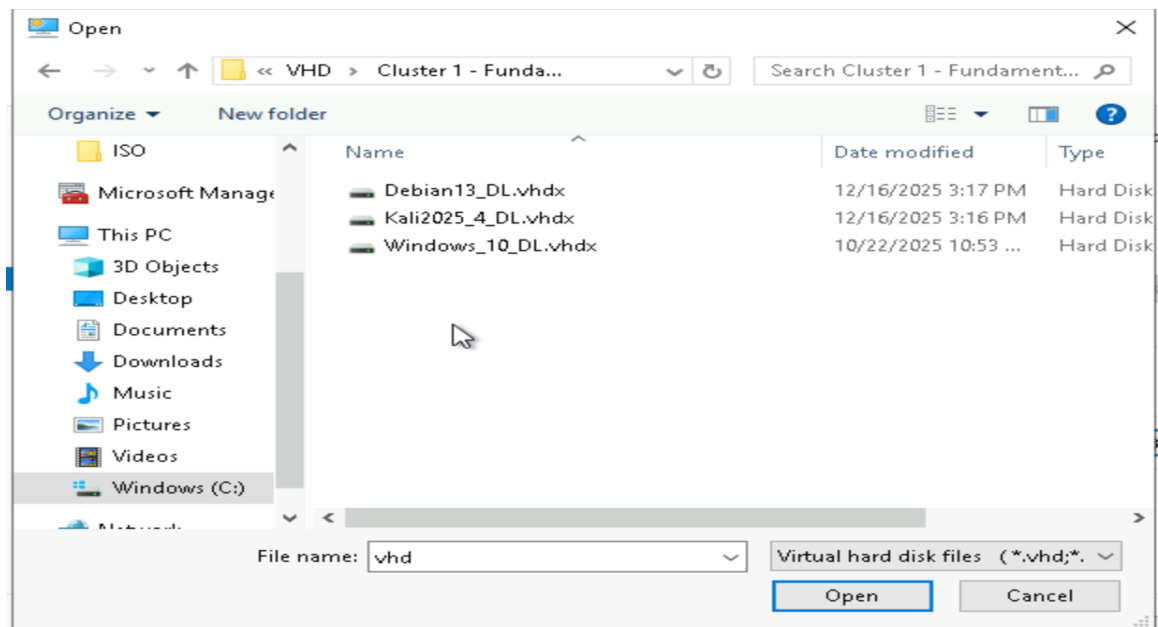
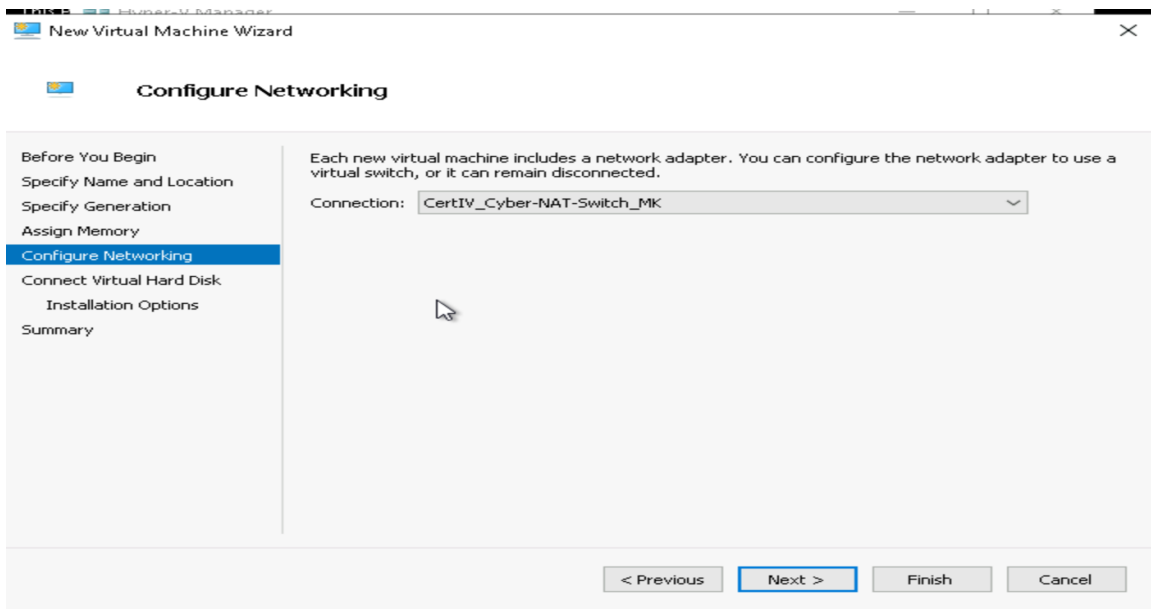
When you decide how much memory to assign to a virtual machine, consider how you intend to use the virtual machine and the operating system that it will run.

< Previous

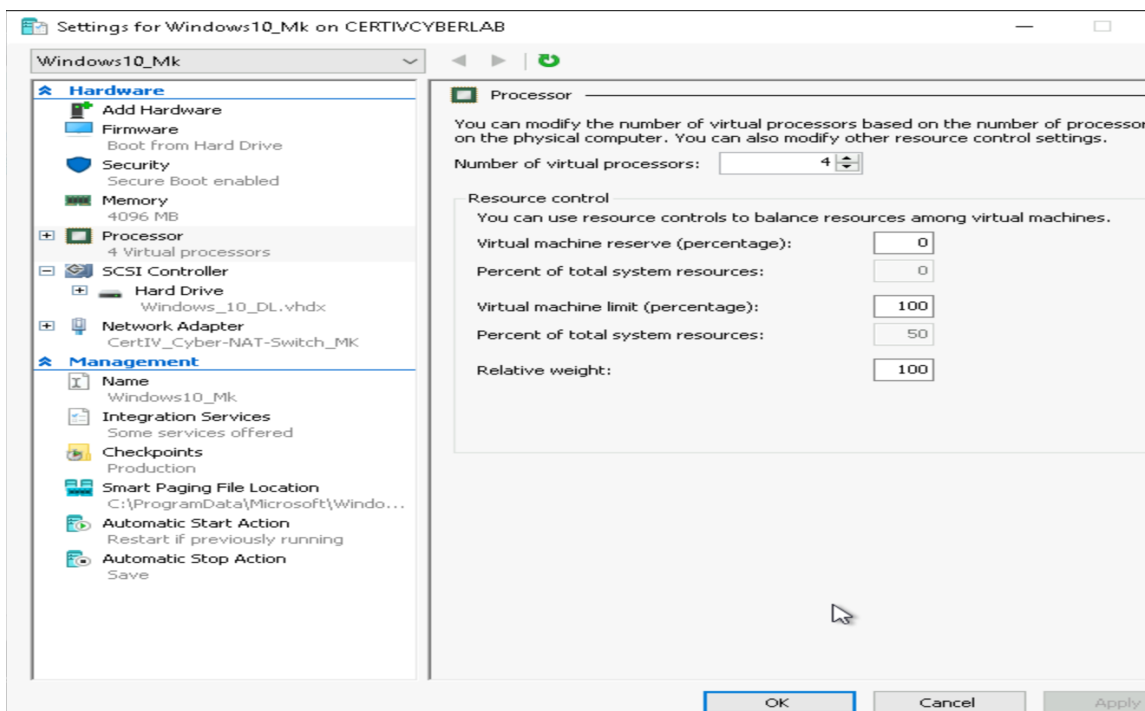
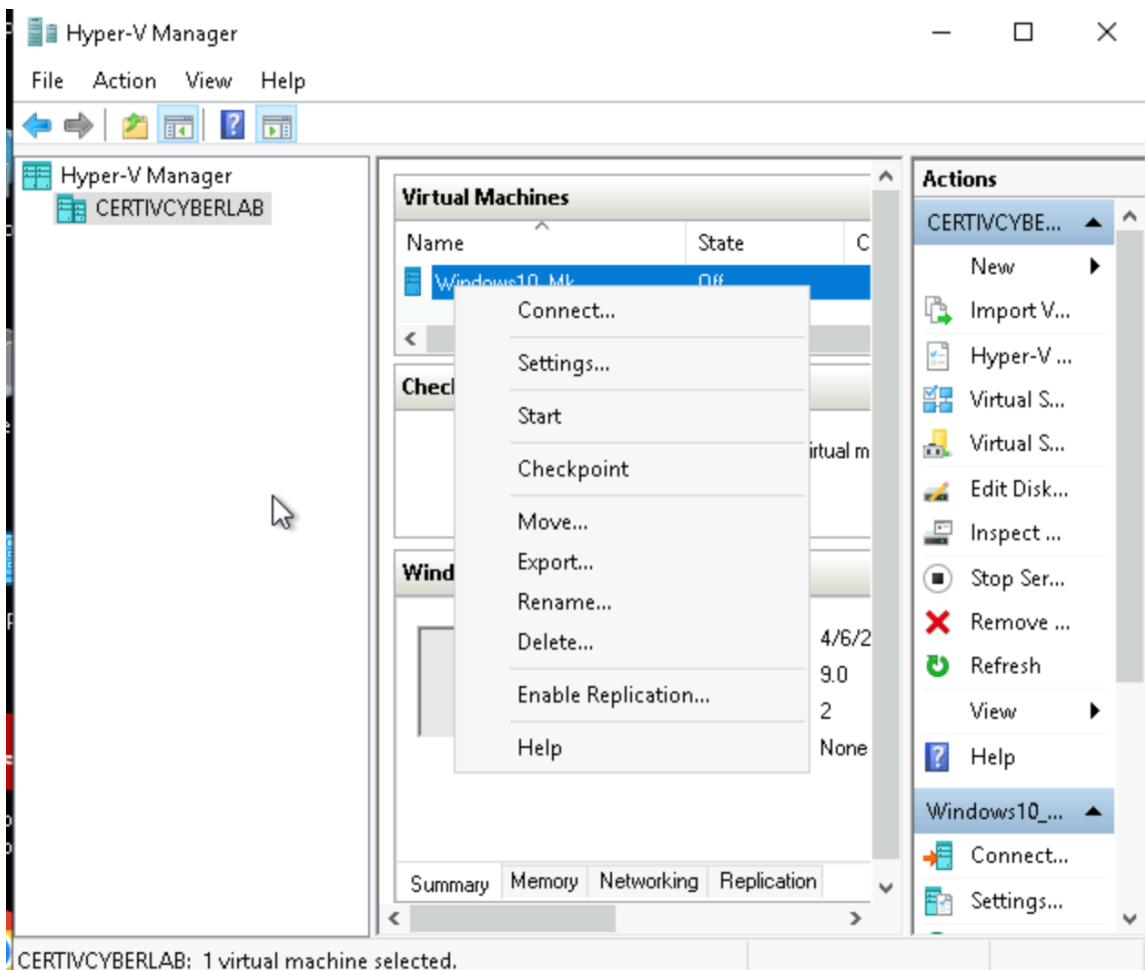
Next >

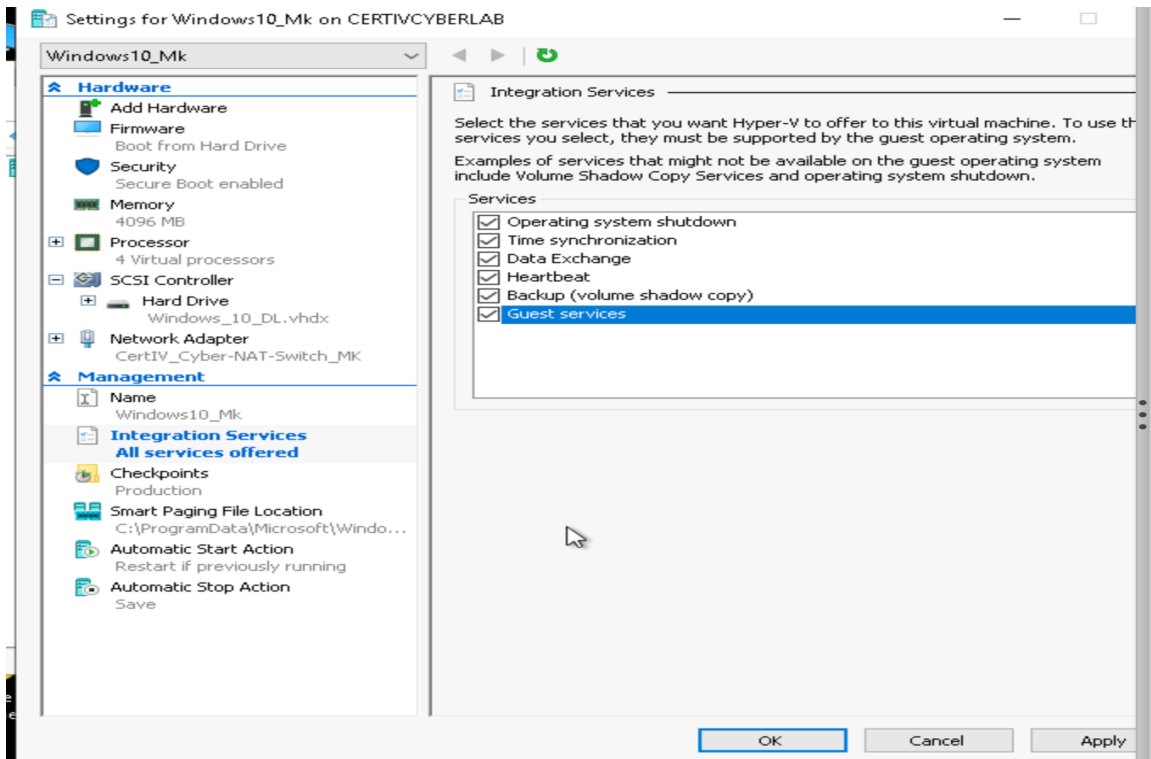
Finish

Cancel

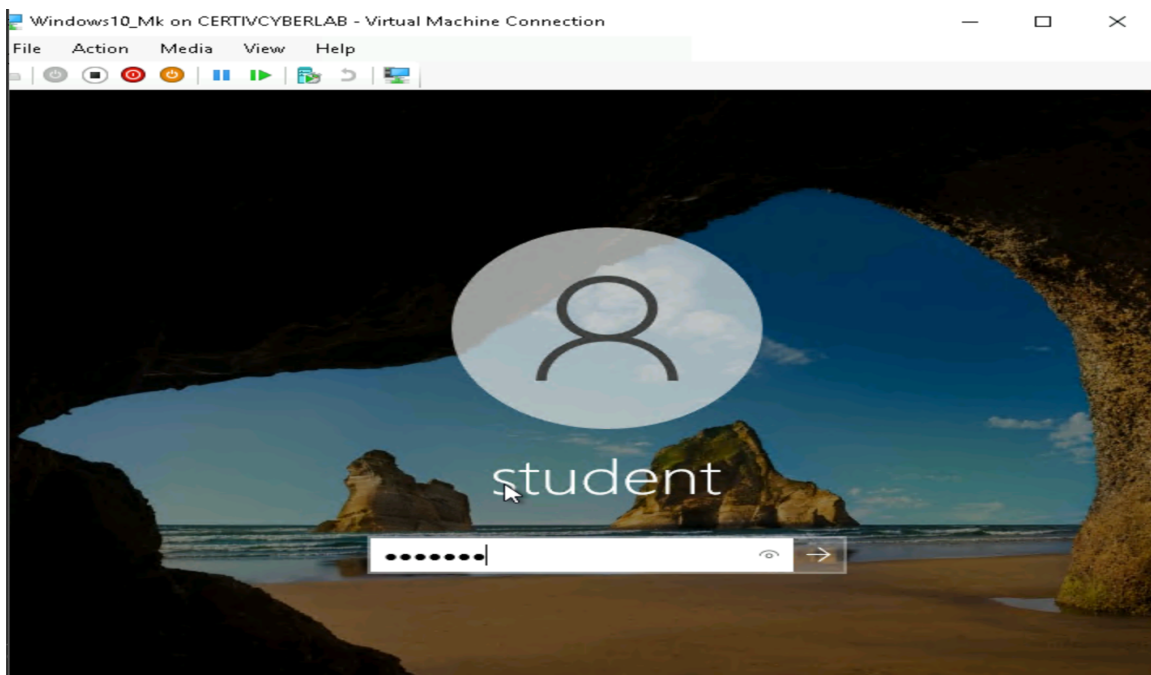


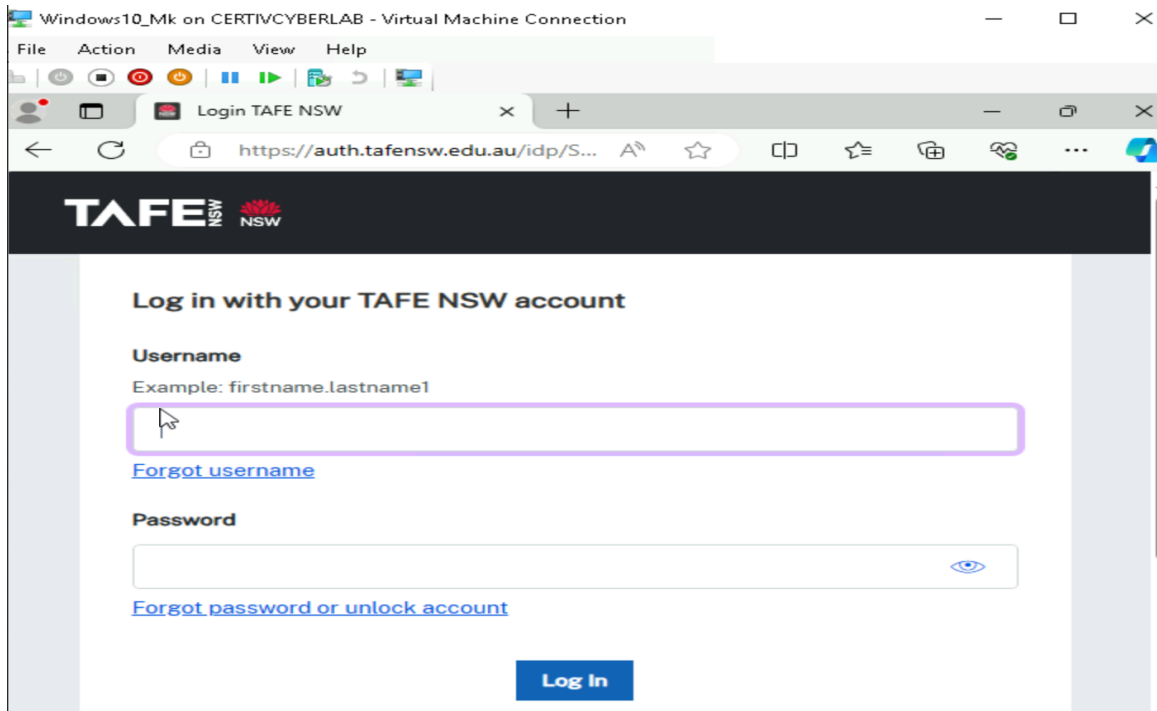
Adjust Processor setting, Right click Windows 10 VM → Settings → Processor → Change to 4 → Apply → Enable guest services, click integration services → check guest services → Apply → OK





Step 2: Connecting to the Nested VM and testing Internet





What This Lab Helped Me Learn

This lab helped me understand how to design and configure a virtual network environment using Hyper-V. I learned how NAT allows virtual machines to access external networks, how DHCP automates IP address allocation, and how Routing and Remote Access integrates these services. Additionally, I gained hands-on experience in troubleshooting network connectivity issues and validating configurations using a nested virtual machine.

Reflection

This lab strengthened my practical system administration and networking skills. It improved my understanding of virtual networking concepts, which are essential in cybersecurity roles such as SOC Analyst and Network Security Engineer.